A lattice model for silicon photomultiplier saturation under arbitrary spatial and temporal light distributions

Author Name
Institution

March 26, 2026

Abstract

Write the abstract here.

1 Introduction

Write the introduction here.

2 Model

2.1 Single-SPAD Response

This section develops a microscopic description of SiPM saturation for arbitrary spatial and temporal illumination profiles. We begin with the response of a single SPAD, then build a lattice model over the full device, take the continuum limit, and finally recover worked examples for uniform, Gaussian, and slit illumination.

For a SiPM with SPADs indexed by (i, j), the probability that the (i, j)-th SPAD avalanches during a single observation window is

$$\rho_{ij} = 1 - \exp(-\lambda_{ij}) \tag{1}$$
$$= 1 - \exp(-QE_{ij}P_{ij}FF_{ij}U_{ij}), \tag{2}$$

where λ_{ij} is the expected number of detected photoelectrons in that SPAD, QE_{ij} is the quantum efficiency, P_{ij} is the avalanche triggering probability, FF_{ij} is the effective fill factor, and U_{ij} is the number of incident photons on that cell during the observation window. The expected number of avalanches from a SiPM with $n_x \times n_y$ SPADs is therefore

$$\langle N \rangle = \sum_{i=1}^{n_x} \sum_{j=1}^{n_y} \left[1 - \exp(-QE_{ij}P_{ij}FF_{ij}U_{ij})\right]. \tag{3}$$

It is convenient to define the intrinsic SPAD photon detection efficiency as

$$PDE_{ij} = QE_{ij}P_{ij}FF_{ij}, \tag{4}$$

so that equation (3) becomes

$$\langle N \rangle = \sum_{i=1}^{n_x} \sum_{j=1}^{n_y} \left[1 - \exp(-PDE_{ij} U_{ij})\right]. \tag{5}$$

The effective device-level photon detection efficiency is then

$$PDE_{\text{eff}} = \frac{\langle N \rangle}{\sum_{i=1}^{n_x} \sum_{j=1}^{n_y} U_{ij}}. \tag{6}$$

If $\lambda_{ij} \ll 1$ for all cells, then the exponential in equation (3) can be linearized to give

$$PDE_{\text{eff}} \approx \frac{1}{\sum_{ij} U_{ij}} \sum_{i=1}^{n_x} \sum_{j=1}^{n_y} PDE_{ij} U_{ij}. \tag{7}$$

Under uniform illumination, $U_{ij} \approx \left(\sum_{ij} U_{ij}\right) / (n_x n_y)$, and therefore

$$PDE_{\text{eff}} \approx \frac{1}{n_x n_y} \sum_{i=1}^{n_x} \sum_{j=1}^{n_y} PDE_{ij}. \tag{8}$$

If the SPAD properties are approximately independent across the array, so that covariance terms are negligible, then the average cell efficiency may be approximated as

$$\langle PDE \rangle \approx \langle QE \rangle \langle P \rangle \langle FF \rangle. \tag{9}$$

In what follows, we refer to equation (9) as the intrinsic PDE of a nominally uniform device, while equation (6) is the effective PDE under a specified illumination pattern.

2.2 Discrete Lattice Model

We now extend the single-SPAD response to a full lattice model over space and time. Let U_{ijk} denote the number of photons incident on cell (i, j) during the k-th reset window. The avalanche probability in that spatiotemporal voxel is

$$\rho_{ijk} = 1 - \exp(-PDE_{ij} U_{ijk}). \tag{10}$$

The expected number of avalanches recorded by a SiPM with $n_x \times n_y$ cells over n_t reset windows is then

$$\langle N \rangle = \sum_{i=1}^{n_x} \sum_{j=1}^{n_y} \sum_{k=1}^{n_t} \left[1 - \exp(-PDE_{ij} U_{ijk})\right]. \tag{11}$$

Equation equation (11) is the general discrete model. The purely spatial two-dimensional lattice is recovered as the special case $n_t = 1$, for which

$$\langle N \rangle = \sum_{i=1}^{n_x} \sum_{j=1}^{n_y} \left[1 - \exp(-PDE_{ij} U_{ij})\right]. \tag{12}$$

2.3 Continuum Limit

If variation in PDE_{ij} across the array is small, we approximate $PDE_{ij} \approx PDE$ and treat the device as a dense lattice with spacings Δx and Δy. Let $\Phi(x, y, t)$ denote the incident photon flux density, in units of photons per unit area per unit time. Then the photons incident on one lattice voxel are

$$U(x, y, t) = \Phi(x, y, t)\, \Delta x\, \Delta y\, \Delta t. \tag{13}$$

Substituting this into equation (11) and taking the dense-lattice limit gives

$$\langle N \rangle = \int \frac{dx\, dy\, dt}{\Delta x\, \Delta y\, \Delta t} \left[1 - \exp(-PDE\, \Phi(x, y, t)\, \Delta x\, \Delta y\, \Delta t)\right]. \tag{14}$$

Equation equation (14) assumes a synchronized-reset time lattice with spacing $\Delta t = \tau$, where τ is the SPAD recovery time. This is a good approximation when the illumination varies slowly compared with τ or when one is interested in steady-state response averaged over many recovery cycles.

For static illumination, the continuum model reduces to

$$\langle N \rangle = \int \frac{dx\, dy}{\Delta x\, \Delta y} \left[1 - \exp(-PDE\, \Phi(x, y)\, \Delta x\, \Delta y)\right]. \tag{15}$$

2.4 Uniform Illumination

The standard uniform-illumination saturation law is recovered immediately as a special case of equation (14). Consider a uniform beam incident on a SiPM with $N_{\text{cells}} = n_x n_y$ cells. If N_{photons} photons are delivered during a single reset window, then each cell receives on average $N_{\text{photons}}/N_{\text{cells}}$ photons and

$$\langle N \rangle = N_{\text{cells}} \left[1 - \exp\left(-PDE \frac{N_{\text{photons}}}{N_{\text{cells}}}\right)\right]. \tag{16}$$

This is the standard SiPM non-linearity formula for uniform illumination [**?**].

For steady illumination with optical power P_{opt}, photon frequency ν, Planck's constant h, and recovery time τ, the incident photon rate is $R_\gamma = P_{\text{opt}}/(h\nu)$. Over one reset window, the mean number of incident photons is $N_{\text{photons}} = R_\gamma \tau$, so the expected avalanche rate is

$$\frac{\langle N \rangle}{T} = \frac{N_{\text{cells}}}{\tau} \left[1 - \exp\left(-PDE \frac{R_\gamma \tau}{N_{\text{cells}}}\right)\right], \tag{17}$$

where $T = n_t \tau$ is the total observation time.

2.5 Gaussian Illumination

Consider a separable two-dimensional Gaussian flux profile,

$$\Phi(x, y) = \frac{N_{\text{photons}}}{2\pi\sigma_x\sigma_y} \exp\left(-\frac{x^2}{2\sigma_x^2} - \frac{y^2}{2\sigma_y^2}\right), \tag{18}$$

where N_{photons} is the number of incident photons per reset window.

Substituting equation (18) into equation (15) yields

$$\langle N \rangle = \int \frac{dx\,dy}{\Delta x\,\Delta y} \left[1 - \exp\left(-PDE\,\frac{N_{\text{photons}}}{2\pi\sigma_x\sigma_y} e^{-x^2/(2\sigma_x^2) - y^2/(2\sigma_y^2)} \Delta x\,\Delta y \right) \right]. \tag{19}$$

Assuming the SiPM active area is large compared with the beam width, so that the Gaussian may be treated as defined over an infinite plane with $\sigma_x, \sigma_y \ll L_{\text{SiPM}}/2$, the integral can be evaluated in closed form as

$$\langle N \rangle = \frac{2\pi\sigma_x\sigma_y}{\Delta x \Delta y} \left[\gamma + \ln\left(\frac{PDE\,N_{\text{photons}}\,\Delta x \Delta y}{2\pi\sigma_x\sigma_y} \right) - \text{Ei}\left(-\frac{PDE\,N_{\text{photons}}\,\Delta x \Delta y}{2\pi\sigma_x\sigma_y} \right) \right], \tag{20}$$

where γ is the Euler-Mascheroni constant and Ei is the exponential integral.

2.6 Slit Illumination

For a slit-like profile that is uniform along y and confined to width w along x, the flux may be written as

$$\Phi(x, y) = \begin{cases} \dfrac{N_{\text{photons}}}{wL_y}, & |x| \le w/2, \\ 0, & |x| > w/2, \end{cases} \tag{21}$$

where $L_y = n_y \Delta y$ is the illuminated extent along the slit direction. The continuum model then gives

$$\langle N \rangle = \frac{wL_y}{\Delta x \Delta y} \left[1 - \exp\left(-PDE \frac{N_{\text{photons}} \Delta x \Delta y}{wL_y} \right) \right]. \tag{22}$$

This result makes explicit how the same total photon number leads to a stronger local saturation when the illumination is concentrated into a narrower region of the device.

3 Monte Carlo

Describe the Monte Carlo model and simulation parameters here.

4 Experiment

Describe the experimental setup and measurement procedure here.

5 Results

5.1 Analytic vs. Monte Carlo

Compare the analytic model and Monte Carlo here.

5.2 Analytic vs. Monte Carlo vs. Experiment

Compare all three approaches for the Gaussian case here.

5.3 Discussion of τ_{eff}

Discuss the fitted effective recovery time here.

6 Conclusion

Summarize the main result, validation, implications, and future work here.

Acknowledgments

Write acknowledgments here if needed.